SAMPLE LESSON CONTENT

Physical and Chemical Properties of Matter: Useful and Harmful Materials

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Matter can be described using two broad categories of properties: physical properties and chemical properties. Understanding the difference between these two types of properties helps explain why some materials are useful in daily life, while others must be handled with caution.

Physical Properties

A physical property is a characteristic of a material that can be observed or measured without changing what the material actually is. Common physical properties include color, shape, size, texture, odor, and the temperature at which a substance melts or boils.

Because physical properties can be observed directly, they are often the first clues used to describe or identify a material. For example, gold can be identified partly by its yellow color and shine, while ice can be identified by its solid form and cold temperature.

Chemical Properties

A chemical property describes how a material behaves when it interacts or reacts with another substance, forming a new material with different properties in the process. Chemical properties cannot always be seen right away — they usually become evident only when a material undergoes a change.

Common examples of chemical properties include flammability, which describes whether a material can catch fire; reactivity, which describes how a material behaves when combined with water, air, or another chemical; and the tendency of some materials to rust or spoil over time.

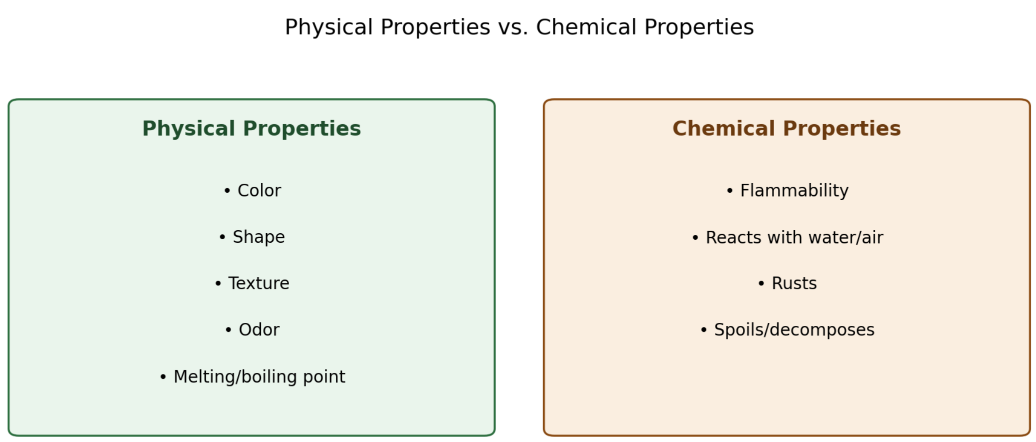

Figure 3. Physical properties can be observed directly, while chemical properties are seen only when a material reacts or changes.

Materials That Are Useful

Many materials are valued in daily life because of the specific properties they have. Cooking oil is useful for frying food because it can withstand high temperatures without burning quickly. Baking soda is useful in cooking because of its chemical property of reacting with acidic ingredients to produce gas bubbles, which helps batter rise.

Materials That Can Be Harmful

Some materials, despite being useful, can also be harmful if they are not handled properly. Bleach and other strong cleaning chemicals can irritate the skin and eyes, and can release harmful fumes if mixed with certain other substances. Flammable materials, such as alcohol or gasoline, can easily catch fire if kept near an open flame or a heat source.

This is why product labels often include warning symbols and instructions — they communicate important information about a material's properties so that it can be used safely and stored properly, especially away from young children.